course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
- The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered
- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours).**

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:

Text Books

1) Brain -Computer Interfacing: An Introduction by Rajesh P. N Rao University of Washington DATE PUBLISHED: September 2013:ISBN:
2) Brain-Computer Interfaces : Foundations and methods Maureen Clerc, Laurent Bougrain, Fabien Lotte

Reference Books

1) Brain-Computer Interfaces 2: Technology and Applications, Volume 2 Maureen Clerc, Laurent Bougrain, Fabien Lotte John Wiley & Sons, 29-Aug-2016 – Computers Schalk, G., & Mellinger, J. (2010).
2) A Practical Guide to Brain–Computer Interfacing with BCI2000: General-Purpose Software for Brain-Computer Interface Research, Data Acquisition, Stimulus Presentation, and Brain Monitoring. Springer Science & Business

Web links and Video Lectures (e-Resources):

- https://sccn.ucsd.edu/wiki/Introduction_To_Modern_Brain- ComputerInterfaceDesign
- https://www.udemy.com/course/brain-computer-interface/

Activity Based Learning (Suggested Activities in Class)/ Practical Based Learning

- Quizzes,
- Assignments
- Seminars

Semester | 7

VII Semester

Biomechanics and Biodynamics			Semester	VII
Course and Course Code	IPCC	**BBM701**	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	3:0:2:0		SEE Marks	50
Total Hours of Pedagogy	40 hours Theory + 8-10 Lab slots		Total Marks	100
Credits	04		Exam Hours	3
Examination nature (SEE)	Theory with Practical			

Course objectives:
After completion of the course, the students will be able to
- Identify, analyze, and solve various biomechanical problems.
- Demonstrate an understanding of kinetic concepts including inertia, force, torque, and impulse.
- Identify the major factors involved in the angular kinematics of human movement.
- Understand human Gait.

Teaching-Learning Process (General Instructions)
These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes.
- Always start every class hour with preamble of what was covered in previous class and what would be discussed in the present class.
- Encourage group discussions and arrange debates on selected topics.
- Try to arrange some industrial visits to understand various process automation techniques.
- Give exhaustive assignments on all topics so that students will be able to practice answering any questions in the University examinations that would come from nook and corner of the syllabus.
- Arrange seminars by the students on certain intriguing topics relevant to syllabus by the students.

MODULE – 1

Biomechanics Applications to Joint Structure and Function: Introduction to Kinematics, Displacement in space, Force vectors and gravity, Linear forces and concurrent forces. Kinetics of rotary and translatory forces. Classes of levers. Close chain force analysis.
Constitutive Equations: Equations for Stress and Strain, Non-viscous fluids, Newtonian viscous fluids, Elastic solids. Visco-elasticity andits applications in biology.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 2

Joint Structure and Function: Properties of connective tissues; Human Joint design; Joint Function and changes in disease.
Integrated Functions: Kinetics and Kinematics of Postures; Static and Dynamic Postures; Analysis of Standing, Sitting and Lying Postures.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 3

Gait Analysis: Gait cycle and joint motion; Ground reaction forces; Trunk and upper extremity motion; internal and external forces, moments and conventions; Gait measurements and analysis.
Force Platform and Kinematic Analysis: Design of force platforms, Integrating force and Kinematic data; linked segment, free-body analysis.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 4	
Bio-Viscoelastic Fluid: Viscoelasticity, Viscoelastic Models: Maxwell, Voigt and Kelvin Models Response to harmonic variation. Use of viscoelastic models. Bio-Viscoelastic fluids: Protoplasm. Mucus, saliva, semen, synovial fluids.	
Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 5	
Rheology of Blood in Microvessels: Fahreus·Lindqulsteffect and inverse effect, hematocrit in very narrow tube. **Finite Element Analysis in Biomechanics:** Model creation, Solution, Validation of results and applications of FEA.	
Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

PRACTICAL COMPONENT OF IPCC*(May cover all / major modules)*

Sl. No	List of experiments to be performed
1	Measurement of EMG Signal using Bio-Pac -Acquisition System
2	Measurement of EEG Signal using Bio-Pac -Acquisition System
3	EEG signal Acquisition using ENO-BIO software
4	EEG signal measurement using Neuro-feedback and Bio-feedback
5	To Read and Plot ECG data with Random noise
6	To Read and Plot ECG data with 50Hz sinusoidal noise
7	Signal Averaging method for a given data
8	Data compression using Turning Point Algorithm using C and Matlab
9	To Read and Plot EEG data, Power Spectrum of EEG
10	Linear Regression
11	Multiple regression
12	Random Forest Classification

Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

- Describe the architecture, functioning and applications of PLC in automation.
- Develop ladder diagram programs for automation systems using different PLC instruction sets
- Analyze the basics of distributed control system and communication protocols used in automation industries.
- Develop process automation system using SCADA and DCS and also develop models of process automation using modern tools.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

CIE for the theory component of the IPCC (maximum marks, 50)

- IPCC means practical portion integrated with the theory of the course.
- CIE marks for the theory component are **25 marks** and that for the practical component is **25 marks**.
- 25 marks for the theory component are split into **15 marks** for two Internal Assessment Tests (Two Tests, each of 15 Marks with 01-hour duration, are to be conducted) and **10 marks** for other assessment methods mentioned in 22OB4.2. The first test at the end of 40-50% coverage of the syllabus and the second test after covering 85-90% of the syllabus.
- Scaled-down marks of the sum of two tests and other assessment methods will be CIE marks for the theory component of IPCC (that is for **25 marks)**.
- The student has to secure 40% of 25 marks to qualify in the CIE of the theory component of IPCC.

CIE for the practical component of the IPCC

- **15 marks** for the conduction of the experiment and preparation of laboratory record, and **10 marks** for the test to be conducted after the completion of all the laboratory sessions.
- On completion of every experiment/program in the laboratory, the students shall be evaluated including viva-voce and marks shall be awarded on the same day.
- The CIE marks awarded in the case of the Practical component shall be based on the continuous evaluation of the laboratory report. Each experiment report can be evaluated for 10 marks. Marks of all experiments' write-ups are added and scaled down to **15 marks**.
- The laboratory test **(duration 02/03 hours)** after completion of all the experiments shall be conducted for 50 marks and scaled down to **10 marks.**
- Scaled-down marks of write-up evaluations and tests added will be CIE marks for the laboratory component of IPCC for **25 marks**.
- The student has to secure 40% of 25 marks to qualify in the CIE of the practical component of the IPCC.

SEE for IPCC

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours**)

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored by the student shall be proportionally scaled down to 50 Marks

The theory portion of the IPCC shall be for both CIE and SEE, whereas the practical portion will have a CIE component only. Questions mentioned in the SEE paper may include questions from the practical component.

Suggested Learning Resources:

1) Introduction to Programmable Logic Controllers, Garry Dunning, 3rdedition, Centage Learning. (Modules: 1, 2 & 3).
2) Computer based Industrial Control, Krishna Kant, 2nd edition, PHI, 2017 (Modules: 4 & 5).
3) Programmable Logic Controllers, F.D. Petruzella, Tata Mc-Graw Hill, Third edition, 2010.
4) Programmable Controllers, T.A. Hughes, Fourth edition, ISA press, 2005.
5) Practical Modern SCADA Protocols: DNP3, 60870.5 and Related Systems, Gordon Clarke, Deon Reynders, Edwin Wright, Newnes, 1st Edition, 2004.

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Quizzes,
- Assignments,
- Seminars

ARM Processor			Semester	VI
Course and Course Code	IPCC	**BBM702**	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	3:0:2:0		SEE Marks	50
Total Hours of Pedagogy	40 hours		Total Marks	100
Credits	4		Exam Hours	3
Examination nature (SEE)	Theory with Practical			

Course objectives:
After completion of the course, the students will be able to

- Understand the basic design and architecture of ARM processor
- To learn the ARM instruction for assembly and **C-**program
- To learn the thumb instruction for assembly and **C-**program and **C** basics for ARM
- Understand the usage of exceptions and interrupts in ARM and operating systems for ARM
- To learn the basic concepts of memory hierarchy, usage of cache memory and memory management

Teaching-Learning Process (General Instructions)
These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes.

- Lecture method(L)does not mean only traditional lecture method, but different type of teaching methods like PPT presentation through LCD maybe adopted to develop the outcomes.
- Show Video/animation films to explain evolution of arm processor development technologies.
- Encourage collaborative (Group)Learning in the class
- Ask atleast three HOTS(Higher order Thinking)questions in the class ,which promotes critical thinking
- Adopt Problem Based Learning (PBL), which fosters students' Analytical skills, develop thinking skills such as the ability to evaluate, generalize, and analyse information rather than simply recall it.
- Showthedifferentwaystosolvethesameprogramtaskandencouragethestudentstocomeupwiththeirown creative ways to solve them.
- Discuss how every concept can be applied to the real world, and when that's possible, it helps improve the students understanding.

MODULE – 1

ARM Embedded Systems: Introduction, RISC design philosophy, ARM design philosophy, Embedded system hardware-AMBA bus protocol, ARM bus technology, Memory, Peripherals, Embedded system software–Initialization(BOOT)code, Operating System, Applications.
ARM Processor Fundamentals: ARM core dataflow model, registers, current program status register, Pipeline, Exceptions, Interrupts and Vector Table.
Text Book 1: Chapter 1: 1.1 to 1.4, Chapter 2: 2.1 to 2.5

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2

MODULE – 2

Introduction to the ARM Instruction set: Introduction, Data processing instructions, Branch instructions, Load-Store instruction, Software interrupt instructions, Program status register instructions, Loading constants, ARMv5E extensions, Conditional Execution
Text Book 1: Chapter 1: 1.1 to 1.4, Chapter 2: 1.1 to 2.5

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2
MODULE – 3	
Introduction to the THUMB instruction set: Introduction, THUMB register usage, ARM–THUMB interworking, Other Branch instructions, Data processing instructions, single-register and multiple-register load-store instructions, Stack instructions, Software interrupt instructions. Efficient C-Programming: Overview of C-Compilers and optimization, Basic C Datatypes, C looping Structures Register allocation, Function calls.	
Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2
MODULE – 4	
Exception and Interrupt Handling: Exception Handling-ARM Processor Exceptions and Modes, Vector Table, Exception Priorities, Link Register Offset, Interrupts-Interrupt Latency, Basic Interrupt Stack design and implementation, Interrupt Handling Scheme- Non nested Interrupt Handler, Nested Interrupt Handler, Reentrant Interrupt Handler, Prioritized Simple Interrupt Handler, **Embedded Operating Systems**: Fundamental Components, SLOS Directory Layout, Initialization, Interrupts and Exceptions handling, scheduler, Context Switch, Device Driver Framework.	
Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L2, L3
MODULE – 5	
CACHES: The memory Hierarchy and caches memory-caches and memory management units, Cache Architecture basic architecture of caches memory, basic operation of cache controller, the relationship between cache and main memory. **Memory Management Units**: Moving from an MPU to an MMU, Virtual memory Working-Defining regions using pagers, multitasking and the MMU, Memory organization in a virtual memory system, page tables Translational look aside buffer	
Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2

Practical Component of IPCC *(May cover all / major modules)*

Sl. No	List of experiments to be performed
1	Write an ALP to find the sum of first 10 integer numbers.
2	Write an ALP to find factorial of a number.
3	Write an ALP to add an array of 16- bit numbers and store the 32- bit result in internal RAM
4	Write an ALP to find the square of a number (1 to 10) using look-up table.
5	Write an ALP to find the largest/smallest number in an array of 32 numbers.
6	Write an ALP to arrange a series of 32 bit numbers in ascending/descending order using Bubble sort algorithm
7	Interface with LPC1768 ARM to Display “Hello World” message using Internal UART.
8	Interface a DAC and generate Triangular and Square waveforms.
9	Interface a Stepper motor and rotate it in clockwise and anti-clockwise direction
10	Interface and Control a DC Motor.
	Demonstration Experiments (for CIE)
11	Interface a 4x4 keyboard and display the key code on an LCD.

12	Demonstrate the use of an external interrupt to toggle an LED ON/OFF.
13	Interface a simple Switch and display its status through Relay, Buzzer and LED
14	Interface a 4x4 keyboard and display the key code on an LCD.

Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

1) After studying this course, students will be able to Depict the organization, architecture, bus technology, memory and operation of the ARM microprocessors.
2) Employ the knowledge of Instruction set of ARM processors to develop basic Assembly Language Programs.
3) Recognize the importance of the Thumb mode of operation of ARM processors and develop C-programs for ARM processors.
4) Describe the techniques involved in Exception and Interrupt handling in ARM Processors and understand the fundamental concepts of Embedded Operating Systems
5) Develop embedded C-programs to interact with Builtin Peripherals for hardware programs,
6) Design, analyse and write programs using Keil software,

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

CIE for the theory component of the IPCC (maximum marks, 50)

- IPCC means practical portion integrated with the theory of the course.
- CIE marks for the theory component are **25 marks** and that for the practical component is **25 marks**.
- 25 marks for the theory component are split into **15 marks** for two Internal Assessment Tests (Two Tests, each of 15 Marks with 01-hour duration, are to be conducted) and **10 marks** for other assessment methods mentioned in 22OB4.2. The first test at the end of 40-50% coverage of the syllabus and the second test after covering 85-90% of the syllabus.
- Scaled-down marks of the sum of two tests and other assessment methods will be CIE marks for the theory component of IPCC (that is for **25 marks)**.
- The student has to secure 40% of 25 marks to qualify in the CIE of the theory component of IPCC.

CIE for the practical component of the IPCC

- **15 marks** for the conduction of the experiment and preparation of laboratory record, and **10 marks** for the test to be conducted after the completion of all the laboratory sessions.
- On completion of every experiment/program in the laboratory, the students shall be evaluated including viva-voce and marks shall be awarded on the same day.
- The CIE marks awarded in the case of the Practical component shall be based on the continuous evaluation of the laboratory report. Each experiment report can be evaluated for 10 marks. Marks of all experiments' write-ups are added and scaled down to **15 marks**.
- The laboratory test **(duration 02/03 hours)** after completion of all the experiments shall be conducted for 50 marks and scaled down to **10 marks.**
- Scaled-down marks of write-up evaluations and tests added will be CIE marks for the laboratory component of IPCC for **25 marks**.

- The student has to secure 40% of 25 marks to qualify in the CIE of the practical component of the IPCC.

Semester End Examination for IPCC

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours**)

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored by the student shall be proportionally scaled down to 50 Marks

The theory portion of the IPCC shall be for both CIE and SEE, whereas the practical portion will have a CIE component only. Questions mentioned in the SEE paper may include questions from the practical component.

Suggested Learning Resources:

Text Books

1) Andrew N Sloss, Dominic System and Chris Wright, "ARM System Developers Guide", Elsevier, Morgan Kaufman publisher, 1stEdition, 2008, ISBN:1758608745

Reference Books

1) David Seal, "ARM Architecture Reference Manual", Addison- Wesley, 2nd Edition, 2009, ISBN:978-0201737196.
2) Furber S, "ARM System on chip Architecture", Addison Wiley, 2nd Edition 2008, ISBN:978-0201675191
3) Rajkamal, "Embedded System", Tata McGraw-Hill Publishers, 2nd Edition, 2008,

Web links and Video Lectures (e-Resources):

- VTU e-shikshana programmes
- VTU Edu-sat programmes
- https://nptel.ac.in/courses/117106111
- https://www.youtube.com/watch?v=4VRtujwa_b8
- https://www.digimat.in/nptel/courses/video/117106111/L30.html

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Quizzes,
- Assignments,
- Seminars
- Mini project

Biometric System			Semester	VII
Course and Course Code	PCC	**BBM703**	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	4:0:0:0		SEE Marks	50
Total Hours of Pedagogy	40 hours		Total Marks	100
Credits	4		Exam Hours	3
Examination nature (SEE)	Theory			

Course objectives:
After completion of the course, the students will be able to

- To introduce the general principles of design of biometric systems and the underlying trade-offs. personal privacy and security implications of biometrics based identification technology. Introduction to fingerprint biometric
- To familiarize with Face recognition and Hand Geometry, feature extraction, pattern classification. Authentication Methods and their algorithms
- To acquire knowledge about various parameters involved in Iris and Voice recognition. Authentication Methods and their algorithms.
- Introduction to multimodal Biometric system and its functional blocks and futuristic biometric systems

Teaching-Learning Process (General Instructions)
These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes.

- Always start every class hour with preamble of what was covered in previous class and what would be discussed in the present class.
- Encourage group discussions and arrange debates on selected topics.
- Give exhaustive assignments on all topics so that students will be able to practice answering any questions in the University examinations that would come from nook and corner of the syllabus.
- Arrange seminars by the students on certain intriguing topics relevant to syllabus by the students.

MODULE – 1

Introduction to biometrics: Introduction and back ground, Biometric technologies, Passive biometrics, Active biometrics, Biometric systems, Enrolment, Templates, Algorithm, Verification, Biometric applications, biometric characteristics- Authentication technologies –Need for strong authentication, Protecting privacy and biometrics and policy, Biometric applications, Biometric characteristics

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 2

Fingerprint Technology: History of fingerprint pattern recognition, General description of fingerprints, Finger print feature processing techniques, Fingerprint sensors using RF imaging techniques – Fingerprint quality assessment, Computer enhancement and modeling of fingerprint images, Fingerprint enhancement– Feature extraction, Fingerprint classification, Fingerprint matching

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 3

Face recognition and Hand Geometry: Introduction to face recognition _ Neural networks for face recognition, face recognition from correspondence maps, Hand geometry, Scanning, Feature Extraction, Adaptive Classifiers, Visual-Based Feature Extraction and Pattern Classification, Feature extraction, Types of algorithm, Biometric fusion.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 4

Iris, Voice recognition: Iris scan, Features, Components, Operation (Steps), Competing iris Scan technologies, Strength and weakness. Voice Scan, Features, Components, Operation (Steps), Competing voice Scan (facial) technologies–Strength and weakness.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 5

Biometric Authentication: Introduction, Biometric Authentication Methods, Biometric Authentication Systems, Biometric authentication by fingerprint -Biometric Authentication by Face Recognition, Expectation, Maximization theory, Support Vector Machines. Biometric authentication by fingerprint, Biometric authentication by hand geometry- Securing and trusting a Biometric transaction, Matching location, local host, authentication server, Match On Card (MOC), Multi-Biometrics and Two-Factor authentication

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

1) Demonstrate knowledge engineering principles underlying biometric systems.
2) Describe and explain Finger print feature processing and techniques, computer enhancement and modelling.
3) Face recognition, how to perform Feature Extraction, classification of features, training of algorithm using neural network
4) Competing iris Scan technologies, various steps involved in voice scan, challenges related to iris and voice scan
5) Demonstration of innovative multimodal Biometric system and Statistical Measures of Biometrics

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
- The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered
- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours).**

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:

1) Arun A. Ross, Karthik Nandakumar, Anil K. Jain, "Introduction to Biometrics", 2011, 1st edition, Springer, New York, USA.
2) Haizhou Li, Liyuan Li, Kar-Ann Toh, Advanced Topics in Biometrics, 2012, 1st edition, World Scientific Publisher, Singapore**.**

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Demonstration of Biometric devices
- Quizzes,
- Assignments,
- Seminars

Professional Elective Course

Biostatistics			Semester	VII
Course and Course Code	PEC	**BBM714A**	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	3:0:0:0		SEE Marks	50
Total Hours of Pedagogy	40 hours		Total Marks	100
Credits	3		Exam Hours	3
Examination nature (SEE)	Theory			

Course objectives:
After completion of the course, the students will be able to

- Understand and apply statistical methods for the design of biomedical analysis.
- Understand and use mathematical and statistical theory underlying the application of biostatistical methods
- Apply knowledge in statistical data analysis.
- Participate in a research team in the development and evaluation of new and existing statistical methodology.

Teaching-Learning Process (General Instructions)
These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes.

- Always start every class hour with preamble of what was covered in previous class and what would be discussed in the present class.
- Encourage group discussions and arrange debates on selected topics.
- Give exhaustive assignments on all topics so that students will be able to practice answering any questions in the University examinations that would come from nook and corner of the syllabus.
- Arrange seminars by the students on certain intriguing topics relevant to syllabus by the students.
- Try to arrange some industrial visit to understand various process automation techniques.

MODULE – 1

Getting Acquainted With Biostatistics: Introduction, Some Basic Concepts, Measurement and Measurement Scales, Sampling and Statistical Inference, The Scientific Method and The Design of Experiments, Computers and Bio Statistical Analysis. (Text Book 1 : Chapter 1)

Strategies For Understanding The Meanings Of Data: Introduction, The Ordered array, Grouped Data : The Frequency Distribution, Descriptive Statistics: Measure of Central Tendency, Descriptive Statistics : Measure of Dispersion. (**Text Book** 1 : Chapter 2)

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 2

Probability: The Basis Of Statistical Inference: Introduction, Two Views of Probability: Objective and Subjective, Elementary Properties of Probability, Calculating the Probability of an Event. (Text Book 1 : 3.1, 3.2, 3.3, 3.4)

Probabilistic Features Of Certain Data Distributions: Introduction, Probability Distributions of Discrete Variables, The Binomial Distribution, The Poisson Distribution, Continuous Probability Distributions

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 3	
Probabilistic Features Of The Distributions Of Certain Sample Statistics: Introduction, Sampling Distribution, Distribution of the Sample Mean, Distribution of the Difference Between Two Samples Means, Distribution of the Sample Proportion, Distribution of the Difference Between Two Sample Proportions. (T**ext Book** 1 : Chapter 5) **Using Sample Data To Make Estimates About Population Parameters :** Introduction, Confidence Interval for a Population Mean, The *t* Distribution, Confidence Interval for the Difference Between Two Population Means, (**Text Book** 1 : 6.1, 6.2, 6.3, 6.4)	
Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 4	
Using Sample Data To Make Estimates About Population Parameters: Confidence Interval for a Population Proportion, Confidence Interval for the Difference Between Two Population Proportions, Determination of Sample Size for Estimating Means, Determination of Sample Size for Estimating Proportions, Confidence Interval for the Variance of a Normally Distributed Population, Confidence Interval for the Ratio of the Variances of Two Normally Distributed Populations. (**Text Book**1 : 6.5, 6.6, 6.7, 6.8, 6.9, 6.10) **Using Sample Statistics To Test Hypotheses About Population Parameters:** Introduction, Hypotheses Testing : A Single Population Mean. (**Text Book** 1 : 7.1, 7.2)	
Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 5	
Using Sample Statistics To Test Hypotheses About Population Parameters: Hypotheses Testing: The Difference Between Two Population Means, Paired Comparisons, Hypotheses Testing : A Single Population Proportion, Hypotheses Testing : The Difference Between Two Population Proportions, Hypotheses Testing : A Single Population Variance, Hypotheses Testing : The Ratio of Two Population Variances. The Type II Error and the Power of a Test, Determining Sample Size to Control Type II Errors. (**Text Book**1 : 7.3, 7.4, 7.5, 7.6, 7.7, 7.8, 7.9, 7.10)	
Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

1) Describe the basic statistical terms, concepts, procedures and statistical measures.
2) Apply probability concepts and probability distributions for statistical inferences.
3) Apply sampling distribution concepts and estimation procedures for population parameters.
4) Select and apply appropriate hypothesis tests for statistical analysis.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
- The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered
- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours).**

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Textbook:

1) Biostatistics: Basic Concepts and Methodology for the Health Sciences – by Wayne W. Daniel, John Wiley & Sons Publication, 9th Edition, 2009.

Reference Books:

1) Principles of Biostatistics, by Marcello Pagano and KimberleeGauvreu, Thomson Learning Publication, Indian Edition, 2007.
2) Biostatistics, by Ronald N Forthofer, EunSul Lee and M. Hernandez, Academic Press, 2007.
3) Basic Biostatistics and its Applications, by Animesh K. Dutta, 2006.

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Quizzes,
- Assignments,
- Seminars

Medical Information and Expert Systems			Semester	VII
Course and Course Code	PEC	**BBM714B**	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	3:0:0:0		SEE Marks	50
Total Hours of Pedagogy	40 hours		Total Marks	100
Credits	3		Exam Hours	3
Examination nature (SEE)	Theory			

Course objectives:
After completion of the course, the students will be able to

- Understand fundamental characteristics of data, information, and knowledge in the Health Informatics domain
- Become familiar with common algorithms for health applications and IT components in representative clinical processes
- Develop understanding of various aspects of Health Information Technology standards
- Become familiar with IT aspects of clinical process modeling and health information systems

Teaching-Learning Process (General Instructions)
These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes.

- Always start every class hour with preamble of what was covered in previous class and what would be discussed in the present class.
- Encourage group discussions and arrange debates on selected topics.
- Give exhaustive assignments on all topics so that students will be able to practice answering any questions in the University examinations that would come from nook and corner of the syllabus.
- Arrange seminars by the students on certain intriguing topics relevant to syllabus by the students.

MODULE – 1

Medical Informatics: Aim and scope, salient feature, Introduction, history, definition of medical informatics, bioinformatics, online learning, introduction to health informatics, prospectus of medical informatics.

Hospital Management And Information Science: Introduction, HMIS: need, Benefits, capabilities, development, functional areas. Modules forming HMIS, HMIS and Internet, Pre-requisites for HMIS- client server technology, PACS, why HMIS fails, health information system, disaster management plans, advantages of HMIS.

Text1: (Section I, 1 and 2, Section II-3)

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 2

Hospital Management And Information Systems-Structure And Functions : Central Registration Module, OPD / Consultant Clinic / Polyclinic Module, Indoor Ward Module, Patient Care Module, Procedure Module, Diet Planning Module, MLC Register Module, Pathology Laboratory Module, Blood Bank Module, Operation Theatre Module, Medical Stores Module, Pharmacy Module, Radiology Module, Medical Records Index Module, Administration Module, Personal Registration Module, Employee Information Module, Financial modules, Health & Family Welfare, Medical Examination, Account Billing, Medical Research, Communication, General Information.

Text 1: (Section II-6)

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 3

Computer Assisted Medical Education: CAME, Educational software, Simulation, Virtual Reality, Teleeducation, Tele-mentoring.

Computer Assisted Patient Education: CAPE, patient counseling software. Computer assisted surgery (CAS), Limitations of conventional surgery, 3D navigation system, intra-operative imaging for 3D navigation system, merits and demerits of CAS.

Text1: (Section III – 7 & 8)

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 4

Telecommunication Based Systems: Tele-Medicine, Need, Advantages, Technology- Materials and Methods, Internet Tele-Medicine, Applications.

Tele-Surgery: Tele-surgery, Robotic surgery, Need for Tele-Surgery, Advantages, Applications.

Text1: (Section V- 13 & 14)

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 5

Knowledge Based And Expert Systems: Introduction, Artificial Intelligence, Expert systems, need for Expert Systems, materials and methods- knowledge representation & its methods, production rule systems, algorithmic method, OAV, object oriented knowledge, database comparisons, statistical pattern classification, decision analysis, tools, neural networks, advantages of ES, applications of ES.

Text 1: (Section II – 4)

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

1) Explain the basics and importance of medical informatics in hospital management.
2) Describe the different modalities functions exist in the hospital for effective management.
3) Explain the role of technology both hardware & software in training the medical personalities.
4) Discuss the role of telecommunication, tele-surgery, robotics in healthcare.
5) Explain the decision making concepts used in healthcare and their applications.
6) Apply information and communication technology in healthcare.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
- The first test will be administered after 40-50% of the syllabus has been covered, and the second test

will be administered after 85-90% of the syllabus has been covered

- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours).**

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:

Textbook:

1) Medical Informatics: A Primer, by Mohan Bansal, 1st Print, Tata McGraw Hill, Publications, 2003.

Reference Books:

1) Medical Informatics: Computer Applications in Health Care and Biomedicine by E.H.Shortliffe, G. Wiederhold, L.E.Perreault and L.M.Fagan, 2ndEdition, Springer Verlag, 2000.
2) Handbook of Medical Informatics by J.H.VanBemmel, Stanford University Press/ Springer, 2000.

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Quizzes,
- Assignments,
- Seminars

Physiological System Modelling			Semester	VII
Course and Course Code	PEC	**BBM714C**	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	3:0:0:0		SEE Marks	50
Total Hours of Pedagogy	40 hours		Total Marks	100
Credits	03		Exam Hours	3
Examination nature (SEE)	Theory			

Course objectives:
After completion of the course, the students will be able to
- To introduce the basic system concepts and differences between engineering and physiological control systems.
- To acquaint students with different mathematical techniques in analysing a system and the various types of nonlinear modelling approaches.
- To teach neuronal membrane dynamics and to understand the procedures for testing, validation and interpretation of physiological models.
- To study the cardiovascular model and apply the modelling methods to multi-input and multi-output systems.

Teaching-Learning Process (General Instructions)
These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes.
- Always start every class hour with preamble of what was covered in previous class and what would be discussed in the present class.
- Encourage group discussions and arrange debates on selected topics.
- Give exhaustive assignments on all topics so that students will be able to practice answering any questions in the University examinations that would come from nook and corner of the syllabus.
- Arrange seminars by the students on certain intriguing topics relevant to syllabus by the students.
- Ask at least three HOTS (Higher-order Thinking) questions in the class, which promotes critical thinking.

MODULE – 1

The problem of system modelling in physiology, Need for modelling, Conceptual and mathematical models – Modelling, experiments and Simulation, Feedback control systems, Difference between engineering and physiological control systems.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 2

Deductive and Inductive modelling, Characteristics of a reliable physiological model, Modelling a simple reflex, and Mathematical modelling.
System Identification, Model Specification, and Model estimation. Types of nonlinear modelling approaches. Non parametric modeling. Volterra and Wiener models. Volterra Kernels. Modelling the vertebrate retina. Analysis of estimation errors.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 3

A general model of the nerve membrane, Action potential and synaptic dynamics, Functional integration in the single neuron, Neuronal systems with point process inputs, Conduction in nerve fibres, Voltage clamp

experiment, Hodgkin Huxley (H-H) model, Circuit analog of the H-H nerve membrane model.	
Teaching-Learning Process **RBT Levels**	Chalk and talk method, YouTube Videos, PowerPoint Presentation. L1, L2, L3
MODULE – 4	
System characteristics, System parameters, System functional properties, Input characteristics, Experimental considerations, Data preparation, Data consolidation, Model specification and estimation tasks, Model validation and interpretation.	
Teaching-Learning Process **RBT Levels**	Chalk and talk method, YouTube Videos, PowerPoint Presentation. L1, L2, L3
MODULE – 5	
Cardiovascular systemic and pulmonary circulation, Lumped model of the cardiovascular system, Pulmonary physiology, Respiratory control system. Modeling of multi input/ multi output systems -The Two-input case, Applications of Two-input modeling to physiological systems.	
Teaching-Learning Process **RBT Levels**	Chalk and talk method, YouTube Videos, PowerPoint Presentation. L1, L2, L3

Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

1) Understand the basic system concepts and differences between an engineering and physiological control systems.
2) Apply different mathematical techniques to analyze a system.
3) Comprehend the various nonlinear modelling approaches.
4) Understand the neuronal membrane dynamics.
5) Apply the procedures for testing, validation and interpretation of physiological models.
6) Comprehend the cardiovascular model.
7) Analyse the modelling methods to multi input and multi output systems.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
- The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered
- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

The Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours).**

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:

Text Books

1) Michael C.K. Khoo, "Physiological Control Systems: Analysis, Simulation and Estimation," 2011, 1st edition, Prentice Hall of India, New Delhi.

Reference Books

1) Suresh Devasahayam, "Signal Processing and Physiological Systems Modeling", 2013, 1st edition, Springer, New York.
2) Joseph D. Bronzino and Donald R. Peterson, "The Biomedical Engineering Handbook", 2015, 4th edition, CRC Press, Florida.

Activity Based Learning (Suggested Activities in Class)/ Practical Based Learning

- Quizzes,
- Assignments,
- Seminars

Advanced Clinical Instrumentation			Semester	VII
Course and Course Code	PEC	**BBM714D**	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	3:0:0:0		SEE Marks	50
Total Hours of Pedagogy	40 hours		Total Marks	100
Credits	3		Exam Hours	3
Examination nature (SEE)	Theory			

Course objectives:
After completion of the course, the students will be able to

- To provide the fundamental knowledge of Clinical Instrumentation, the science associated with the measurement of biological variables such as pressure, temperature etc related to human body, the complexities associated with the measurement of the biological parameters and the care that are to be taken for the measurement since it is concerned with human life
- Understanding basic principles and phenomena in the area of advanced clinical instrumentation,
- Theoretical and practical preparation enabling students to maintain medical instrumentation.

Teaching-Learning Process (General Instructions)
These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes.

- Always start every class hour with preamble of what was covered in previous class and what would be discussed in the present class.
- Encourage group discussions and arrange debates on selected topics.
- Give exhaustive assignments on all topics so that students will be able to practice answering any questions in the University examinations that would come from nook and corner of the syllabus.
- Arrange seminars by the students on certain intriguing topics relevant to syllabus by the students.

MODULE – 1

ICU Equipment's and Neonatal Equipment: Oxygen concentrators – Capnographs monitoring systems – cardiac monitor, multipara monitor, Advanced defibrillators –internal and external – Intermediate level of suction apparatus – Laryngoscope, Advance level of radiant warmer, phototherapy units, Doppler fetal heart rate device (handheld type), Fetal Tocography, Baby Incubator, Neonatal ventilator

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 2

Diagnostic Equipment's: Stereo toxic unit- depth recording system-dot scanners- transcutaneous nerve Stimulator- anesthesia monitor, EEG controlled anesthesia- bio-feedback equipments, Spinal reflex measurements. Basic Blood gas analyzer, Photometer and spectrophotometer, Microtome, osometer, Lab freezer, PH meter, Optical microscope, Water bath types, Centrifuge (table), Shakers, Lab, laminar air flow units – Lab precision balances, Pippets, Washers, Incubator and Heating unit centrifuge (Flour) – Electrophoresis systems, tissue embedding equipment, Ambulance setup.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 3

Surgical Equipment's: Warmer (Blood and Patient), tourniquet, insufflators, irrigation unit – Operating microscope, arthroscopic, Operation Theater (OT): Lights, and Patient's tables, Flow meters (gas & blood), sterilizing units (auto clave), Surgical driller, Sterilizing producers, manifold unit – Central supply of air. Laparoscope, Gastro scope, endoscopes -light sources. Bronchoscope: Video processors, Camera, and

Fiber optic cable. Physiological effects of stimulation, galvanic, Faradic and surged types, interferential therapy.	
Teaching-Learning Process **RBT Levels**	Chalk and talk method, YouTube Videos, PowerPoint Presentation. L1, L2, L3
MODULE – 4	
Fundamental Troubleshooting Procedures: Making of an Electronic Equipment, causes of Equipment Failure, Troubleshooting Process & Fault finding Aids, Troubleshooting Techniques, Grounding Systems in Electronic Equipment, Temperature Sensitive Intermittent Problems, and Correction Action to repair the Equipment.	
Teaching-Learning Process **RBT Levels**	Chalk and talk method, YouTube Videos, PowerPoint Presentation. L1, L2, L3
MODULE – 5	
Biomedical Equipment Troubleshooting: Trouble shooting of ECG Machine, EEG Machine, Defibrillator Electrosurgical unit, Anaesthesia machine, Autoclaves & sterilizers, Endoscope. Troubleshooting of Incubators, Nebulizer, Oxygen Concentrators, Oxygen cylinders & flow meters, Pulse Oximeter, Sphygmomanometers, Suction Machine, X-Ray Machine Troubleshooting.	
Teaching-Learning Process **RBT Levels**	Chalk and talk method, YouTube Videos, PowerPoint Presentation. L1, L2, L3

Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

1) Describe about various neonatal and ICU equipment's
2) Discuss the use of surgical equipment's
3) Analyze different types Diagnostic devices
4) Understand fundamental troubleshooting procedures for biomedical instruments
5) Analyze different types troubleshooting techniques.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
- The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered
- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours).**

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:

Textbooks

1) Albert M, Cook and Webster J G, "Therapeutic Medical Devices", Prentice Hall Inc., New Jersey, 1982.
2) Geddes L A and Baker L E, "Principles of Applied Biomedical Instrumentation", John Wiley, 3rd Edition,
3) 1975, Reprint 1989.
4) Khandpur R S, "Hand-book of Biomedical Instrumentation", Tata McGraw Hill, 2nd Edition, 2003.
5) Khandpur R S, "Troubleshooting Electronic Equipment- Includes Repair & Maintenance", Tata McGraw-Hill, Second Edition 2009.
6) Dan Tomal & Neal Widmer, "Electronic Troubleshooting", McGraw Hill, 3rd Edn.

Reference Books

1) Leslie Cromwell, Fred J, Weibell, Erich A, Pfeiffer, "Biomedical Instrumentation andMeasurements", Prentice-Hall India, 2nd Edition,1997.
2) John G, Webster, "Medical Instrumentation application and design", JohnWiley, 3rd Edition, 1997.
3) Feinberg B N, "Applied Clinical Engineering", Prentice Hall Inc., New Jersey, 1986.
4) Nicholas Cram & Selby Holder, "Basic Electronic Troubleshooting for Biomedical Technicians", TSTC Publishing, 2nd Edition 2010.
5) World Health Organization, "Maintenance & Repair of Laboratory, Diagnostic Imaging & Hospital Equipment", Geneva,1994.
6) Ian R, McClelland , "X-ray Equipment maintenance & repairs workbook for Radiographers & Radiological Technologists", World Health Organization, Geneva, 2004.

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Quizzes,
- Assignments,
- Seminars

Open Elective Course

Neural Network and Fuzzy Logic Systems			Semester	VII
Course and Course Code	PEC	**BBM755A**	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	3:0:0:0		SEE Marks	50
Total Hours of Pedagogy	40 hours		Total Marks	100
Credits	03		Exam Hours	3
Examination nature (SEE)	Theory			

Course objectives:
After completion of the course, the students will be able to

- **Preparation**: To prepare students with fundamental knowledge and comprehensive understanding of artificial neural networks and Fuzzy Logic systems.
- **Core Competence**: To equip students to develop and configure ANNs with different types of learning algorithms and to understand the basics of Fuzzy logic operations and systems for real world problems.
- **Professionalism & Learning Environment**: To inculcate an engineering student an ethical and professional attitude by providing an academic environment inclusive of effective communication, teamwork, ability to relate engineering issues to a broader social context, and life-long learning needed for a successful professional career.

Teaching-Learning Process (General Instructions)
These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes.

- Always start every class hour with preamble of what was covered in previous class and what would be discussed in the present class.
- Encourage group discussions and arrange debates on selected topics.
- Give exhaustive assignments on all topics so that students will be able to practice answering any questions in the University examinations that would come from nook and corner of the syllabus.
- Arrange seminars by the students on certain intriguing topics relevant to syllabus by the students.
- Ask at least three HOTS (Higher-order Thinking) questions in the class, which promotes critical thinking.

MODULE – 1

Introduction: Neural Networks, Application Scope of Neural Networks, Fuzzy Logic, Genetic Algorithm, Hybrid Systems, Soft Computing.

Artificial Neural Network: **An Introduction**, Fundamental Concept, Evolution of Neural Networks, Basic models of Artificial Neural Networks (ANN), Important Technologies of ANNs, McCulloch-Pitts Neuron, Linear Separability.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 2

Supervised Learning Network – Introduction –Perceptron Networks, Adaptive Linear Neuron (Adaline), Multiple Adaptive Linear Neurons, Hebb Network and simple problems.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 3

Back –Propagation Network: Theory, Architecture, Flowchart for training process, Training Algorithm,

Learning Factors of Back-Propagation Network, Testing Algorithm of Back-Propagation Network. Radial Basis Function Network, Time Delay Neural Network, Functional Link Networks, Tree Neural Networks, wavelet neural network.	
Teaching-Learning Process **RBT Levels**	Chalk and talk method, YouTube Videos, PowerPoint Presentation. L1, L2, L3
MODULE – 4	
Introduction to Fuzzy Logic, Classical sets and Fuzzy sets: Introduction to Fuzzy Logic, Classical sets (crisp sets), Operations on Classical sets, Properties of Classical sets, Function of Mapping of Classical sets. Fuzzy sets – Fuzzy set operations, Properties of fuzzy sets. Simple Problems **Classical Relations and Fuzzy Relations**: Introduction, Cartesian Product of Relation, Classical Relation, Fuzzy Relation, Tolerance and Equivalence Relations, Simple Problems.	
Teaching-Learning Process **RBT Levels**	Chalk and talk method, YouTube Videos, PowerPoint Presentation. L1, L2, L3
MODULE – 5	
Membership Functions: Introduction, Features of the Membership functions, Fuzzification, Simple Problems. **Defuzzification**: Introduction, Lamba-cuts for Fuzzy sets (Alpha-Cuts), Lamba-Cuts for Fuzzy Relation, Defuzzification Methods. **Fuzzy Logic Control Systems:** Introduction, Control System Design, Architecture and Operation of FLC system, FLC system Models, Application of FLC systems.	
Teaching-Learning Process **RBT Levels**	Chalk and talk method, YouTube Videos, PowerPoint Presentation. L1, L2, L3

Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

8) Compare and contrast the biological neural network and ANN.
9) Discuss the ANN for pattern classification.
10) Develop and configure ANN's with different types of functions and learning algorithms.
11) Apply ANN for real world problems.
12) Discuss the fundamentals of fuzzy logic, implementation and their functions.
13) Apply fuzzy logic concepts in building automated systems.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
- The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered
- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of

assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours).**

5. The question paper will have ten questions. Each question is set for 20 marks.
6. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
7. The students have to answer 5 full questions, selecting one full question from each module.
8. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:

Text Books

2) S. N. Sivanandam and S.N. Deepa, "Principles of Soft Computing", 2nd Edition, Wiley India Pvt. Ltd.-2014.
3) Timothy J Ross, "Fuzzy logic with engineering applications", McGraw Hill International Edition, 1997

Reference Books

3) Simon Haykin, "Neural Networks: A comprehensive foundation", 2nd Edition, PHI, 1998.

Web links and Video Lectures (e-Resources):

- http://www.nptel.ac.in/courses/106105152/
- https://nptel.ac.in/courses/106/106/106106139

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Quizzes,
- Assignments,
- Seminars

Medical Informatics			Semester	VII
Course and Course Code	OEC	**BBM755B**	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	3:0:0:0		SEE Marks	50
Total Hours of Pedagogy	40 hours		Total Marks	100
Credits	3		Exam Hours	3
Examination nature (SEE)	Theory			

Course objectives:
After completion of the course, the students will be able to

- Making data available: Ensuring data is available quickly and reliably
- Improving patient-physician relationships: Collecting, safeguarding, and understanding health data to positively impact the relationship between patients and physicians
- Providing decision support: Helping clinicians make decisions by processing data and knowledge
- Improving patient care: Helping doctors get a better picture of a patient's situation, determine the best treatment approach, and avoid unnecessary examinations
- Preventing diseases: Helping to recognize and prevent diseases
- Bringing research to patient care: Quickly applying research findings to patient care

Teaching-Learning Process (General Instructions)
These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes.

- Always start every class hour with preamble of what was covered in previous class and what would be discussed in the present class.
- Encourage group discussions and arrange debates on selected topics.
- Give exhaustive assignments on all topics so that students will be able to practice answering any questions in the University examinations that would come from nook and corner of the syllabus.
- Arrange seminars by the students on certain intriguing topics relevant to syllabus by the students.

MODULE – 1

Introduction, history, definition of medical informatics, bio-informatics, online learning, introduction to health informatics, prospectus of medical informatics. Impact of Systems on Health Care, Care Providers and Organizations, mobile health care technologies.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 2

Hospital Management Need for HMIS, Capabilities & Development of HMIS, functional area, modules forming HMIS, (like Pathology Lab, Blood bank, Pharmacy, Diet planning). Maintenance and development of HMIS-Ideal Features and functionality of CPR, Development tools for CPR.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 3

Computer Assisted Medical Education: CAME, Educational software, Simulation, Virtual Reality, Tele-education, Tele-mentoring. Computer Assisted Patient Education: CAPE, patient counseling software. Computer assisted surgery (CAS), Limitations of conventional surgery, 3D navigation system, intra-operative imaging for 3D navigation system, merits and demerits of CAS.

Teaching-Learning Process **RBT Levels**	Chalk and talk method, YouTube Videos, PowerPoint Presentation. L1, L2, L3

MODULE – 4

Telecommunication Based Systems: Tele-Medicine, Need, Advantages, Technology- Materials and Methods, Internet Tele-Medicine, Applications. Tele-surgery, Robotic surgery, Need for Tele-Surgery, Advantages, Applications. Real-time Telemedicine. Data Exchange: Network Configuration, circuit and packet switching, H.320 series (Video phone based ISBN) T.120, H.324. Video Conferencing.

Teaching-Learning Process **RBT Levels**	Chalk and talk method, YouTube Videos, PowerPoint Presentation. L1, L2, L3

MODULE – 5

Knowledge Based and Expert Systems: Introduction, Artificial Intelligence, Expert systems, need for Expert Systems, materials and methods- knowledge representation & its methods, production rule systems, algorithmic method, OAV, object oriented knowledge, database comparisons, statistical pattern classification, decision analysis, tools, neural networks, advantages of ES, applications of ES.

Teaching-Learning Process **RBT Levels**	Chalk and talk method, YouTube Videos, PowerPoint Presentation. L1, L2, L3

Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

1) Explain the basics and importance of medical informatics in hospital management.
2) Describe the different modalities functions exists in the hospital for effective management.
3) Discuss the role of telecommunication, tele-surgery, robotics in healthcare.
4) Explain the decision making concepts used in healthcare and their applications.
5) Apply information and communication technology in healthcare.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
- The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered
- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours).**

1. The question paper will have ten questions. Each full question is set for **20 marks**.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to **50 marks**

Suggested Learning Resources

Text Books

1) Medical Informatics: A Primer - by Mohan Bansal, 1st Print, Tata McGraw Hill, Publications, 2003.
2) A.S. Tanenbaum, "Computer Networks", 2012, 5th Edition, Pearson Education, London.
3) Kenneth R. Ong, "Medical Informatics: An Executive primer", 2015, 1st Edition, HIMSS Publishing, Chicago

Reference Books

1) Medical Informatics: Computer Applications in Health Care and Biomedicine by E.H.Shortliffe, G. Wiederhold, L.E.Perreault and L.M.Fagan, 2ndEdition, Springer Verlag, 2000.
2) Handbook of Medical Informatics by J.H.VanBemmel, Stanford University Press/ Springer, 2000.

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Quizzes,
- Assignments,
- Seminars

Sensors and Measurements			Semester	VII
Course and Course Code	OEC	**BBM755C**	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	3:0:0:0		SEE Marks	50
Total Hours of Pedagogy	40 hours		Total Marks	100
Credits	3		Exam Hours	3
Examination nature (SEE)	Theory			

Course objectives:
After completion of the course, the students will be able to

- To acquaint students basics of measurements system and investigate inductive and magnetic sensors
- Compare the resistive and capacitive sensors
- Interpretation of electromagnetic sensors and self-generating sensors
- Validate signal conditioning of sensors

Teaching-Learning Process (General Instructions)
These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes.

- Always start every class hour with preamble of what was covered in previous class and what would be discussed in the present class.
- Encourage group discussions and arrange debates on selected topics.
- Give exhaustive assignments on all topics so that students will be able to practice answering any questions in the University examinations that would come from nook and corner of the syllabus.
- Arrange seminars by the students on certain intriguing topics relevant to syllabus by the students.
- Try to arrange some industrial visit to understand various Aircraft Instruments.

MODULE – 1

Measurement System – Instrumentation – Classification and Characteristics of Transducers – Static and Dynamic – Errors in Measurements – Calibration – Primary and secondary standards.
Resistive sensors, Potentiometers, Strain gauges, Pressure resistive temperature detectors (RTD), thermistors – Magnetoresistors, Light dependent resistor (LDR), Resistive hygrometers, Resistive gas sensors, Liquid conductivity sensors, Capacitive sensors- Variable capacitor, Differential capacitor.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 2

Inductive sensors, Variable reluctance sensors, Eddy current sensors, Linear Variable Differential Transformers (LVDT), Variable transformers, Magneto-elastic and magnetostrictive sensors, Super conducting quantum interference devices (SQUID). Electromagnetic sensors, Sensors based on Faraday's law, Hall effect sensors.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 3

Thermoelectric sensors, Piezo electric sensors, Pyroelectric sensors, Photovoltaic sensors, Electrochemical sensors.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 4

Deflection bridges – Amplifiers, AC carrier system, Current transmitters, Oscillators and Resonators.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 5

Flow measurement systems: Measurement of velocity, Volume and mass flow rate, Heat transfer effect in measurement systems: Characteristics of thermal sensors, Ultrasonic measurement systems.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

1) Gain the basic idea of measurements and the errors associated with measurement
2) Differentiate between the types of sensors available
3) Select a suitable sensor for a given application
4) Apply the knowledge about the measuring instruments to use them more effectively
5) Relate the self-generating sensors with passive sensors
6) Comprehend the basics of signal conditioning
7) Comprehend the operation and characteristics of special measurement systems

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
- The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered
- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours).**

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.

3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:

Text Books

1) Jacob Fraden, "Handbook of modern sensors", 2016, Stn Edition, Springer International Publishing, Switzerland.

Referance Books

1) Ramon Pallas-Areny and John G. Webster, "Sensors and Signal Conditioning", 2012, 211d Edition, John Wiley and Sons Inc, New Jersey.
2) John. G. Webster and HalitEren, "Measurements, Instrumentation and Sensors Handbook: spatial, mechanical, thermal and radiation measurements", 2014, fd edition, CRC Press, Florida.
3) Winncy Y. Du, "Resistive, Capacitive, Inductive, and Magnetic Sensor Technologies", 2015. 1st Edition, CRC Press Taylor & Francis Group, New York.

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Quizzes,
- Assignments,
- Seminars

Biomedical Image Processing			Semester	VII
Course and Course Code	OEC	**BEI755D**	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	2:2:0:0		SEE Marks	50
Total Hours of Pedagogy	40 hours		Total Marks	100
Credits	3		Exam Hours	3
Examination nature (SEE)	Theory			

Course objectives:
After completion of the course, the students will be able to

- To study the image fundamentals and mathematical transforms necessary for image processing.
- To study the image enhancement techniques
- To study image restoration procedures.
- To study the image compression procedures.
- To understand image analysis algorithms.

Teaching-Learning Process (General Instructions)
These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes.

- Always start every class hour with preamble of what was covered in previous class and what would be discussed in the present class.
- Encourage group discussions and arrange debates on selected topics.
- Give exhaustive assignments on all topics so that students will be able to practice answering any questions in the University examinations that would come from nook and corner of the syllabus.
- Arrange seminars by the students on certain intriguing topics relevant to syllabus by the students.

MODULE – 1

Fundamentals of Digital image, Image formation, visual perception, CCD & CMOS Image sensor, Image sampling: Two dimensional Sampling theory, Nonrectangular grid and Hexagonal sampling, Optimal sampling, Image quantization, Non uniform Quantization, Image formats. Types of pixel Operations, Types of neighborhoods, adjacency, connectivity, boundaries, regions, 2D convolution, Color models.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 2

Image Enhancement in Spatial Domain: Background, Point processing – Image negatives, Log transformations, Power law transformations, Contrast stretching, Intensity level slicing, Bit plane slicing, Histogram processing – Histogram equalization, Histogram matching (specification), Arithmetic/Logic operations – Image subtraction, Image averaging. Fundamentals of spatial filtering, Smoothing spatial filters, Sharpening spatial filters.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 3

Image Enhancement In Frequency Domain: Background, 2D-Discrete Fourier Transform and its Inverse, Basic properties of the 2D-Discrete Fourier Transform, Basics of filtering in the frequency domain. Image smoothing using frequency domain filters – Ideal low pass filters, Butterworth low pass filters, Gaussian low pass filters; Image sharpening using frequency domain filters – Ideal high pass filters, Butterworth high pass filters, Gaussian high pass filters, Homo-morphic filtering.

Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

MODULE – 4	
Image Segmentation Detection of discontinuities, Point-line- edge detection, Linear and Circular Hough Transform, Basic Global and Adaptive Thresholding, Region Based segmentation, K-Means Clustering.	
Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3
MODULE – 5	
Image Restoration: Model of the Image degradation/restoration process, Noise models, Restoration using spatial filtering: Mean filters, Order statistic filters - Median filter, Min and Max filters, Midpoint filter. **Image Compression:** Fundamentals, Image compression models, Basic compression methods – Huffman coding, Arithmetic coding, LZW coding, Run-length coding.	
Teaching-Learning Process	Chalk and talk method, YouTube Videos, PowerPoint Presentation.
RBT Levels	L1, L2, L3

Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

1) Define the general terminology of digital image processing.
2) Identify the need for image transforms and their types both in spatial and frequency domain.
3) Identify different types of image enhancement techniques.
4) Describe image segmentation models and learn image segmentation techniques.
5) Explain and apply various methodologies for image compression.

Assessment Details (both CIE and SEE)

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- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours).**

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a

maximum of 3 sub-questions), **should have a mix of topics** under that module.

3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources

Text Books

1) Digital Image Processing - Rafael. C. Gonzalez and Richard. E. Woods, Third Edition, Pearson Education, 2008.

Reference Books

1) Fundamentals of Digital Image Processing - Anil K. Jain, 5th Indian Print, PHI, 2002.
2) Digital Image Processing and Computer Vision - Milan Sonka, India Edition, Cengage Learning.

Activity Based Learning (Suggested Activities in Class)/ Practical Based Learning

- Quizzes,
- Assignments,
- Seminars

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